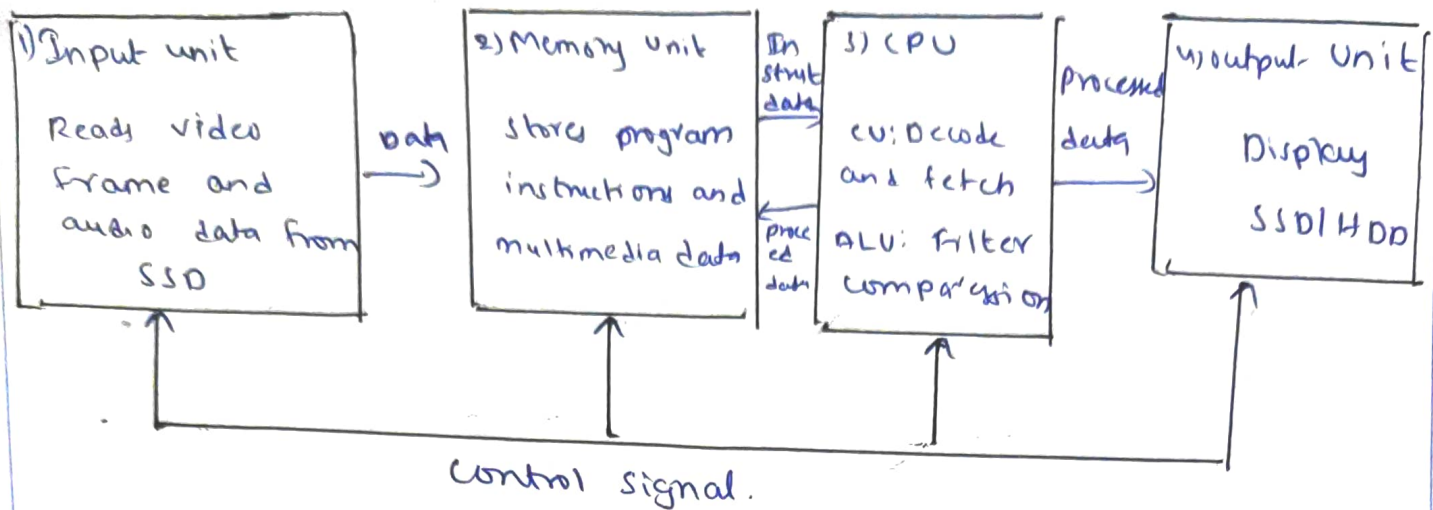


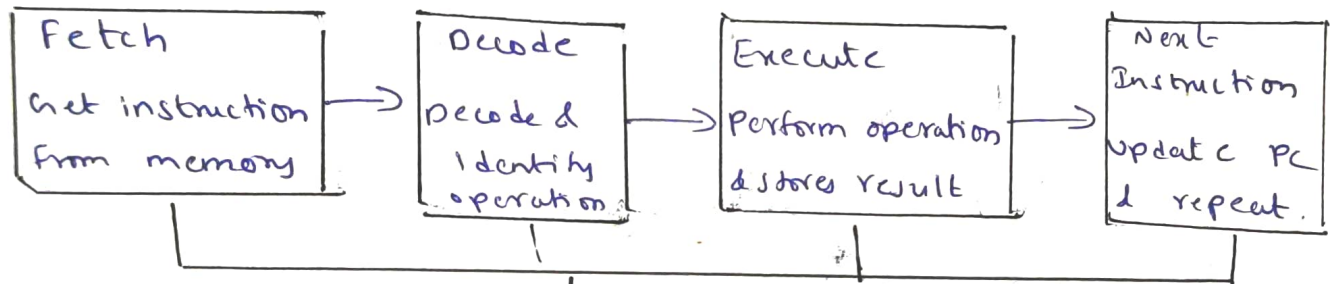
# Video Editing Application



## Working

- 1) The input unit retrieve data from SSD
- 2) Memory stores and provide it to CPU
- 3) CPU processes the data
- 4) Output unit display preview and save final movie.

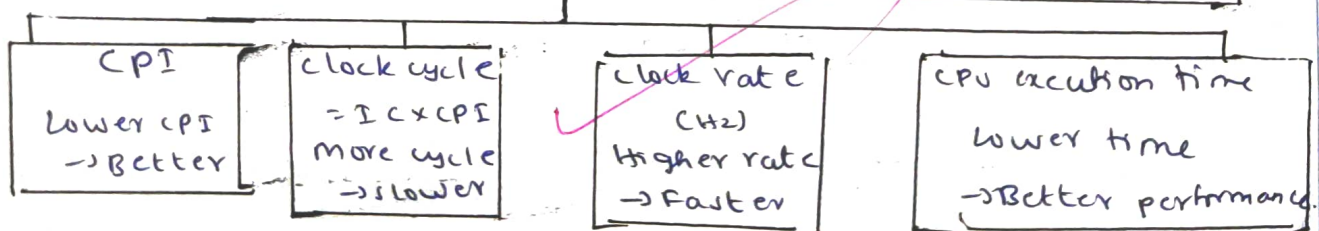
## 2. Instruction Execution & CPU performance



CPU Execution time

$$\text{CPU time} = \text{Instruction count} \times \text{CPI} \times \text{clock cycle time}$$

$$= IC \times CPI \times (1/\text{Clock Rate})$$



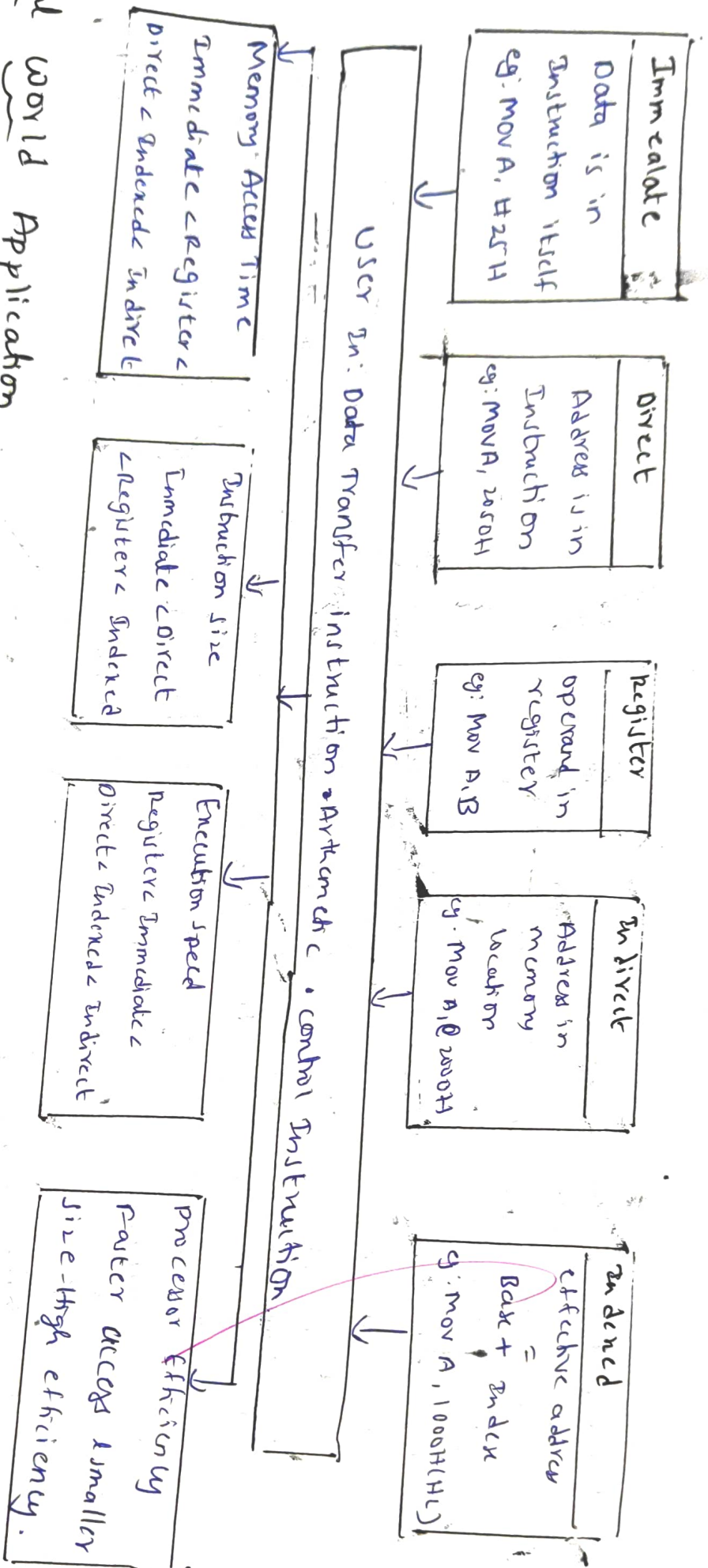
## Improve performance

- ✗ Reduce CPI
- ✗ optimize code & memory
- ✗ Increase clock rate

### 3. Comparison: 8085 vs 8086 Architecture

| Feature             | 8085                                             | 8086                                                                     | Impact on System            |
|---------------------|--------------------------------------------------|--------------------------------------------------------------------------|-----------------------------|
| Architecture        | 8-bit<br>Micro processor                         | 16-bit<br>Micro processor                                                | 8086 → faster execution     |
| Register            | B, C, D, E, H, L<br>(8-bit)                      | AX, BX, CX, DX, SP, BP, SI, DI<br>(16-bit)                               | 8086 → more register        |
| Addressing mode     | 5 modes<br>(Imm, Reg, Direct, Reg indirect, Imp) | 12 modes<br>(All 8085 modes + Displacement, Base Indexed, Base, Indexed) | 8086 → more flexibility     |
| Memory organization | 16-bit Address Bus<br>64KB memory<br>linear      | 20-bit Address Bus<br>1MB memory                                         | 8086 → larger memory        |
| Memory segmentation | No segmentation                                  | yes                                                                      | 8086 → support multitasking |

# 1. Addressing modes & Their Impact.



## Real World Application

- \* Constants
- \* I/O Device
- \* Loop counters
- \* Arrays.

# Binary & Hexadecimal Number System

